

Shubham Singh

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Summary

Software engineer at Amazon with ~4 years of experience building production agentic AI systems, distributed orchestration, and large-scale compliance infrastructure on AWS. Architect and tech lead behind two production multi-agent systems (policy generation + agentic Q&A) and the platform powering 2,000+ compliance control launches in 2026. Strong IC focused on shipping production AI infrastructure.

Experience

Amazon

Aug 2022 – Present

Software Development Engineer II

Seattle, WA

- Architected a **multi-agent policy generation system** (planner → drafter → critic → refiner) using **Strands Agents SDK** on **AWS Bedrock AgentCore runtime**, with stateful human-in-the-loop refinement and LLM-as-judge critique against compliance rubrics; collapsed compliance control authoring from ~**6 months to 2–3 days**.
- Built **MaverickIQ**, an agentic Q&A system on Strands Agents SDK + AgentCore, with hybrid retrieval (**OpenSearch BM25** over structured control metadata + **kNN vector search** over unstructured regulatory corpora), secure tool-calling, and citation-grounded responses with full audit logging; deflected **90%** of manual control-interpretation tickets across **5 partner teams**.
- Designed and led implementation of **DISCO-TA**, a **Step Functions + DynamoDB** orchestration platform for compliance testing automation; replaced brittle Lambda chains with stateful workflows using Map states for per-marketplace parallelism and callback patterns for long-running operations; cut workflow failure rate from **8% to <1%** with full state recoverability and replay.
- Led cross-team design** of **ZeroTouch**, Amazon's internal control launch platform (React/TypeScript + AWS); **sole author of the system HLD**, coordinating **3 partner teams across 3 timezones**; powering **2,000+ compliance controls** launching in 2026 and collapsing control onboarding from weeks of engineering effort to self-serve hours.
- Led design of a **PySpark-based EU-DSA compliance pipeline** processing **30M+ monthly records** with schema versioning, validation, and error isolation; reduced incident resolution time by **70%**.
- On-call across **5 production services** on a rotating cadence; mentored a new-grad SDE through full ramp-up to independent on-call readiness, including pairing on incident triage and authoring design reviews for their feature work.

Open Source & Projects

Microsoft MarkItDown Contributor (Python, Email Parsing, Markdown) – added EML email conversion support for Microsoft's open-source document-to-Markdown AI tooling. **2026**

Strands Multi-Agent Orchestrator (Python, MCP, FastAPI) – multi-agent routing with MCP tools, FastAPI streaming, and observability tracing. **2026**

RAG Platform (Go, OpenSearch) – multi-tenant RAG using BM25 + kNN hybrid retrieval with async ingestion via Redis Streams. **2025**

Technical Skills

AI/LLM: AWS Bedrock, AgentCore, Strands Agents SDK, RAG, Hybrid Retrieval (BM25 + kNN), Tool-Calling, MCP, Prompt Engineering, LLM-as-Judge

Languages: Java, Python, TypeScript, Go, SQL, C/C++

Cloud & Infrastructure: AWS (Lambda, Step Functions, DynamoDB, S3, API Gateway, CloudFront, WAF, Glue), Docker, Kubernetes, CI/CD, Infrastructure as Code

Data & Backend: PySpark, ETL Pipelines, REST APIs, Microservices, Event-Driven Architecture, OpenSearch

Frontend: React, Redux, TypeScript

Education

Indiana University Bloomington

Jan 2021 – Jul 2022

Masters in Data Science

Bloomington, IN